Examination study companion

Renewable Energy

Theory, diagrams and recall

A concise LaTeX guide to the recurring theory in seven papers

Purpose. Understand a system, draw it from memory, then explain its working in a clear sequence. This guide consolidates **99 theory subparts** from the supplied 2022–2025 papers into **24 topic sheets**. Repeated questions share one explanation; numerical solutions are in the separate numerical booklet.

What to study first

Counts below mean *papers containing at least one theory question in that family*, not a prediction of the next examination. Every family below is covered.

Topic family	Papers	Pages
Energy resources, need and alternatives	7/7	3–4
Solar thermal collectors and applications	7/7	6–7
Biomass, biofuels and conversion systems	7/7	9–12
Wind energy conversion systems	7/7	13–15
Hydroelectric plants and hydraulic turbines	7/7	16–18
Geothermal resources and plants	7/7	19–21
Costs, cash flow and financial appraisal	7/7	25–26
Ocean thermal energy conversion (OTEC)	6/7	22–23
Solar radiation geometry	5/7	5
Wave energy devices	5/7	24
Solar photovoltaic technologies	2/7	8

How the pages are organised

Each topic starts with the idea in plain words, then gives an examination sketch, a working sequence, and the comparisons or merits usually requested. Shaded recall boxes are for *closed-book practice*. The final pages provide a revision drill, a question-to-topic coverage index, and sources.

Drawing convention. All figures are original simplified vector schematics, not to scale. Arrows show the stated energy or material flow. Add a title and labels when redrawing; artistic detail earns less than correct connections.

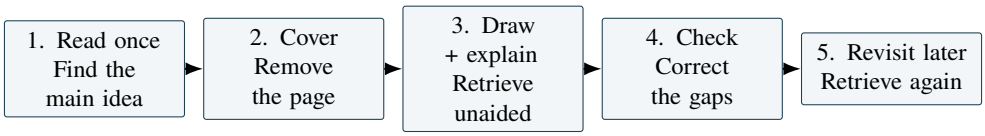
Scope notes. The source calls one biogas plant “floating dome”; the standard floating-drum design is explained. Tar sands and oil shale are included because the paper asks them, but they are *unconventional fossil resources*. Current-energy facts are dated explicitly.

Prepared for memory-focused revision | Compiled with pdfLaTeX | Source crosswalk: p. 28

Study method

How to remember the theory

Use a five-step recall loop



Why this method? In experiments with prose, retrieving material improved later retention more than repeated study, although restudy could look better on an immediate test. Confidence after rereading is therefore an unreliable check of readiness. [M1]

Space the attempts. A practical starting schedule is: learn today, retrieve tomorrow, then after about three and seven days. This is an adaptable example, not a scientifically optimal universal timetable. Research shows that a useful spacing interval depends on how long the information must be retained. Shorten the gap after a weak attempt; lengthen it after an accurate one. [M2]

Draw to explain. Experiments found memory benefits from drawing words rather than only writing them. Applying that idea to engineering sketches is a study strategy: draw the parts, connect the arrows, then explain what each part does. The cited drawing experiments did not directly test engineering examination marks. [M3]

A ten-minute topic routine

Time	Action
0–3 minutes	Read the principle and working. Ask “What enters, what changes, what leaves?”
3–5 minutes	Close the page. Draw and label the system from a blank sheet.
5–7 minutes	Explain the working aloud in four to six steps. Add one reason for each key step.
7–9 minutes	Reopen the page. Correct missing arrows, names and reasons in another colour.
9–10 minutes	Write one comparison and two merits/limitations. Schedule the next recall.

Use the same answer structure in the examination

Define → Draw → Describe → Discuss. Start with a one-sentence definition. Draw a titled, labelled sketch. Number the working steps in the order shown by the arrows. Finish with the requested advantages, limitations or applications. For a classification question, use a small tree or table instead of a long paragraph.

Make a causal chain, not a list. For example: “Pressure is reduced → some hot water flashes into steam → steam drives a turbine.” The word *because* helps: a condenser is needed because it removes rejected heat and supports the low turbine-exhaust pressure.

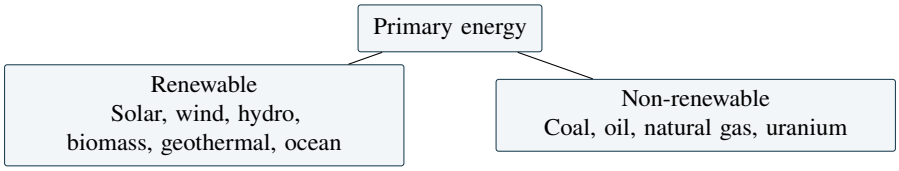
Memory cue. The short hooks in this guide are author-created reminders. Learn the physical meaning first, then use the hook to retrieve it.

Cover and recall. Can you draw the five-step recall loop without looking? In your next attempt, test yourself before reopening the page.

01 | Energy fundamentals

Energy resources and the need for renewables

Core idea. A renewable source is replenished on a human time scale; a non-renewable resource is a finite stock. Electricity is an energy carrier, not a primary resource.



Classification by replenishment. Biomass requires sustainable feedstock management.

Other classifications often asked

Basis	First group	Second group
Stage of conversion	Primary: coal, sunlight, wind	Secondary: electricity, hydrogen, refined fuels
Market transaction	Commercial: purchased electricity, diesel	Non-commercial: gathered fuelwood, crop residues for own use
Established practice	Conventional: historically established locally	Non-conventional: newer alternatives in that context

"Conventional" depends on place and time. Large hydro is renewable even where it is conventional; nuclear is non-renewable despite low operational carbon emissions.

Why India needs renewable energy

1. **Energy security:** locally available resources reduce dependence on imported fuels and fuel-price shocks.
2. **Environment and health:** cleaner generation can reduce air pollution and greenhouse-gas emissions.
3. **Access and development:** distributed systems can supply remote loads, irrigation and local enterprises.
4. **Long-term supply:** renewable flows diversify the energy mix as demand grows and finite resources decline.

Benefits and practical limitations

Solar and wind use no purchased fuel during operation, but their output varies with weather and time. Hydro and some geothermal/biomass plants can provide controllable output, subject to resource and plant constraints. Storage, transmission, flexible demand and a suitable mix of sources support reliable supply.

Renewable projects also have land, materials, ecological and social impacts. Hydro can affect rivers and communities; biomass can compete for land and emit pollutants. "Renewable" does not mean "impact-free".

Global energy crisis: a mismatch between affordable, reliable supply and demand, aggravated by fuel depletion, unequal access, geopolitical disruption, pollution and climate pressures. Responses include efficiency, conservation, diversification and clean supply. [T1]

Memory cue. Stock versus flow. Non-renewables draw down a stock; renewables use a replenished flow.

Cover and recall. Classify uranium, hydrogen, fuelwood and electricity. Give four reasons for renewable energy in India and two limitations.

02 | Energy fundamentals

Alternatives and the current energy picture

Compare alternatives to conventional fossil supply

Option	Main benefit	Main constraint
Solar	Modular electricity/heat; no operating fuel	Day/night and weather variation; land/storage needs
Wind	Low operational emissions; scalable generation	Site-dependent output; grid, wildlife and noise concerns
Hydro	Efficient conversion; storage can offer flexibility	Suitable sites, river impacts, drought and displacement
Biomass	Storable feedstock; can use organic residues	Collection costs, moisture, emissions and sustainable supply
Geothermal	Firm heat and electricity at suitable sites	Exploration/drilling cost; reservoir and fluid management
Nuclear	Firm, low-carbon electricity; high energy density	High capital cost, long development, waste and safety duties

Nuclear working in one line: controlled fission → heat → steam → turbine-generator. Nuclear is an alternative to burning fossil fuels, but uranium is finite, so it is *not renewable*. Renewable and low-carbon are different classifications.

Tar sands and oil shale: wording trap in the paper

Tar/oil sands contain heavy bitumen mixed with sand, water and mineral matter. Mining or in-situ recovery obtains bitumen, which is upgraded/refined into fuels.

Oil shale is rock containing kerogen. Heating (retorting) converts kerogen into vapours that are condensed and processed into shale oil. Oil shale is different from oil already present in a shale formation.

Both can extend petroleum supply, but they are **unconventional fossil resources**, with energy, water, land and emissions burdens. They should not be called renewable merely because a question groups them with "alternatives". [T2]

A dated snapshot for a "present scenario" answer

India: installed capacity, 30 September 2026	Solar **171.05 GW**; wind **59.20 GW**; total renewables including large hydro **299.25 GW**. Non-fossil capacity including nuclear: **308.03 GW**. [D1]
World: electricity generated in 2025	Renewables and coal each supplied approximately **34%**; low-emissions sources (renewables plus nuclear) supplied **43%**. [D2]

How to write the answer: state the date and metric, mention continued fossil dependence alongside renewable growth, then explain the need for efficiency, networks, storage and flexible supply. *Installed capacity* (GW) is a power rating; *generation* (TWh or a share of electricity) is energy delivered over time. Do not compare them as if they were the same metric.

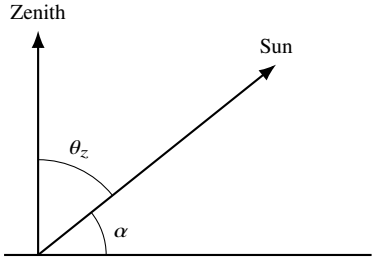
Memory cue. Renewable? Refilled. Low-carbon? Emissions. These ask different questions.

Cover and recall. Explain solar versus nuclear in four points. Why are tar sands not renewable? Recall one dated India figure and one global generation figure.

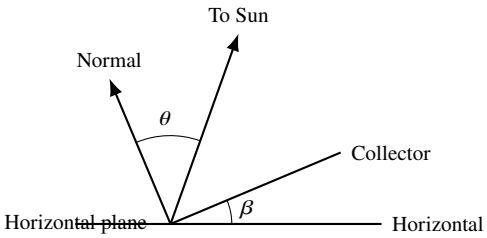
03 | SOLAR ENERGY

Solar radiation geometry

Core idea. Solar angles locate the Sun relative to the Earth, the horizon and the collector. Always state the azimuth and hour-angle sign convention.

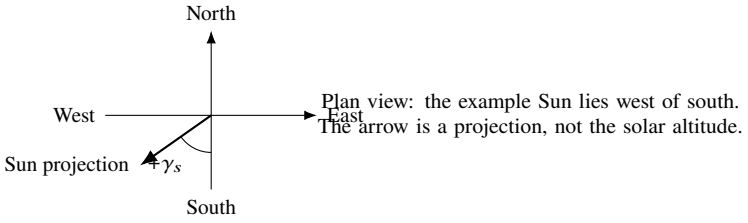


Local vertical section: $\alpha + \theta_z = 90°$



Incidence is measured from the normal

Angle	Definition to write in the examination
Latitude ϕ	Site's angular position north (+) or south (−) of the equator.
Declination δ	Sun–Earth direction's angle to Earth's equatorial plane; varies approximately between −23.45° and +23.45°.
Hour angle ω	Angular displacement from solar noon: 15° per solar hour; morning negative, afternoon positive here.
Altitude α	Angle of the Sun above the local horizontal.
Zenith θ_z	Angle between Sun direction and local vertical; complement of altitude.
Solar azimuth γ_s	Angle of the Sun's horizontal projection from a stated reference. Here: south = 0, west positive, east negative.
Surface tilt β	Angle of collector plane above horizontal.
Surface azimuth γ	Bearing of the collector's outward normal projected horizontally; same south-based convention here.
Incidence θ	Angle between incoming solar direction (toward the Sun) and the collector's outward normal.



Useful relation: $\boxed{\omega = 15°(t_s - 12)}$, where t_s is local solar time in hours. Clock time needs longitude and equation-of-time corrections. [T3]

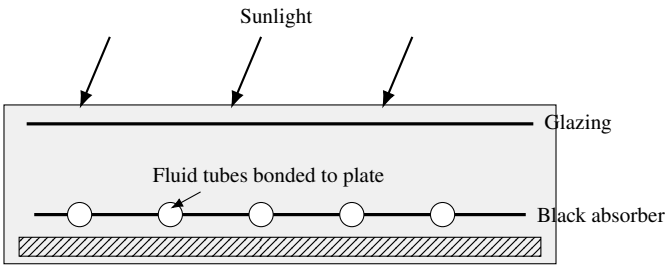
Memory cue. Altitude from floor; zenith from ceiling; incidence from normal.

Cover and recall. Draw altitude and zenith first, then add tilt and incidence. Define any five angles without mixing solar azimuth with surface azimuth.

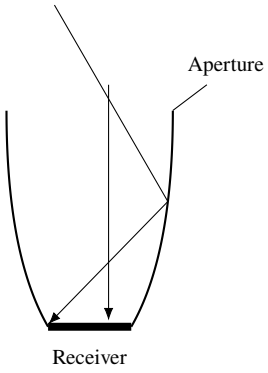
04 | Solar thermal systems

Solar collectors: flat plate and CPC

Classification. Non-concentrating collectors include flat plates and evacuated tubes. Concentrating collectors include line-focus troughs, point-focus dishes/towers, and non-imaging compound parabolic collectors (CPCs). Collectors may also be liquid/air heating, stationary/tracking, and glazed/unglazed.



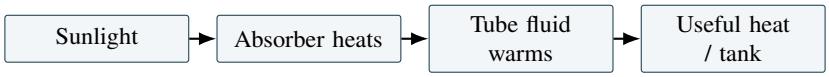
Flat-plate cross-section: casing encloses the assembly



Two facing parabolic segments
CPC section

Liquid flat-plate collector: construction and working

1. Solar radiation passes through a transparent glass cover and reaches a dark/selective absorber plate.
2. The plate absorbs radiation and transfers heat by conduction to bonded tubes.
3. Water or a heat-transfer liquid flows through the tubes and carries heat to a tank or load.
4. The cover reduces convection losses; back/side insulation reduces heat loss through the casing. The cover also affects radiative exchange.



Merits: simple, stationary, accepts direct and diffuse radiation, suitable for water/space heating. **Limits:** heat losses rise at higher fluid temperature; output falls with weak sunlight. It is not a high-concentration device. [T4,T5]

Compound parabolic collector

A CPC uses two shaped reflecting surfaces to direct rays within an *acceptance angle* onto a smaller receiver, without forming a sharp image of the Sun. Low-concentration versions can operate without continuous tracking, sometimes with seasonal adjustment. Higher concentration narrows angular acceptance. A receiver transfers absorbed heat to a fluid; uses include water and process heating. Unlike a simple parabolic trough, a CPC is a *non-imaging* concentrator. [T6]

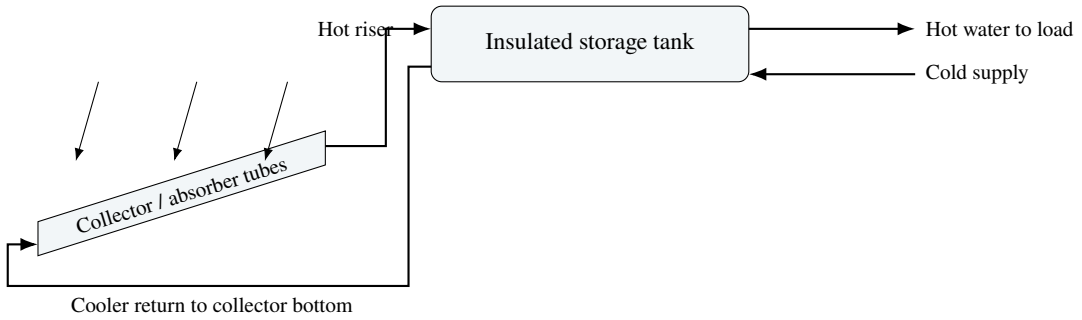
Memory cue. Glass admits; black plate absorbs; tubes carry; insulation saves.

Cover and recall. Draw the four layers of a flat plate. Explain why the cover and insulation have different jobs. Sketch a CPC and identify its receiver.

05 | Solar thermal applications

Solar water heating and drying

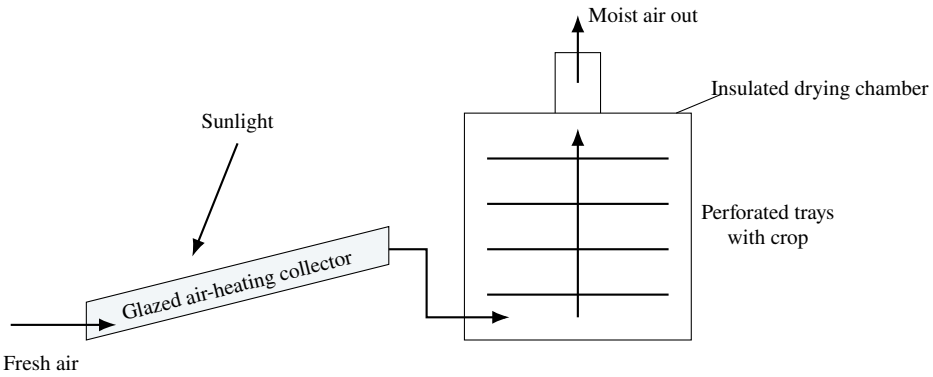
Solar water heater: natural-circulation arrangement



Working: sunlight heats the collector water. Warmer, less-dense water rises to the tank; denser, cooler water returns to the collector. This *thermosiphon* loop needs the tank above the collector and suitable piping. Hot water is drawn from the tank; cold supply replaces it. Pumped systems use a controller and pump instead. An indirect system separates the collector fluid from potable water using a heat exchanger. [T5]

Applications: domestic bathing, hostels, hospitals and low-temperature process heat. **Merits:** saves purchased heating energy; simple operation. **Limits:** variable solar input, storage/backup needs, scaling or freezing depending on water and climate.

Indirect solar dryer: protected product, heated air



Working: the absorber heats incoming air → warm air passes across the crop → heat evaporates crop moisture → moving air carries water vapour out. A chimney provides natural draft; a fan gives forced convection. Moisture removal requires both *heat* and *air exchange*. [T7]

Types: direct dryer exposes the crop to sunlight in a glazed chamber; indirect dryer uses a separate air collector; mixed-mode combines both. Dryers protect produce from dust/insects and can improve control, but depend on weather and correct temperature/airflow. Applications include fruits, vegetables, grains and fish.

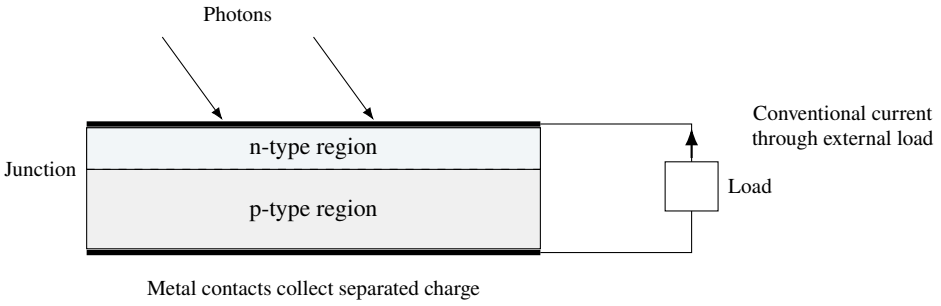
Memory cue. Water heater: hot water rises. Dryer: warm air carries moisture away.

Cover and recall. Redraw both systems. Why is the thermosiphon tank high? What happens if a dryer has heat but no moist-air outlet?

06 | Solar electricity

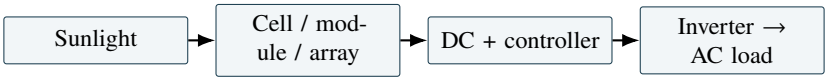
Photovoltaic cells and technologies

Core idea. A photovoltaic (PV) cell converts light directly into electricity in a semiconductor. It does not first boil water or drive a heat engine.



Working in four steps

1. Photons with sufficient energy are absorbed and create electron–hole pairs.
2. The junction's internal electric field helps separate the charge carriers.
3. Contacts collect the carriers; a connected external circuit carries direct current (DC).
4. Cells form modules; modules form arrays. An inverter converts DC to AC when required, while controls match the operating point to conditions.



PV technologies: classify, then compare

Technology	Feature to remember
Monocrystalline silicon	Single-crystal wafers; established high-performance technology.
Multicrystalline silicon	Many crystal grains in a wafer; historically used for lower-cost production, with grain-related losses.
Thin film	Thin semiconductor layers; examples: amorphous silicon, CdTe and CIGS. Material and substrate affect flexibility and performance.
Multi-junction / emerging	Multiple absorbers can use more of the spectrum; perovskite and tandem technologies are active development areas. Commercial maturity varies.

Advantages: modular, no moving parts in the cell, quiet, direct electricity, useful for rooftops, pumps, lighting and remote loads. **Limitations:** night/weather variability, shading losses, temperature effects, conversion equipment, and end-of-life/material management. Batteries are optional storage, not an essential part of every grid-connected system. [T8,T9]

Memory cue. Light creates; junction separates; contacts collect.

Cover and recall. Draw a labelled junction and the external load. Explain why PV produces DC and name three cell-technology families.

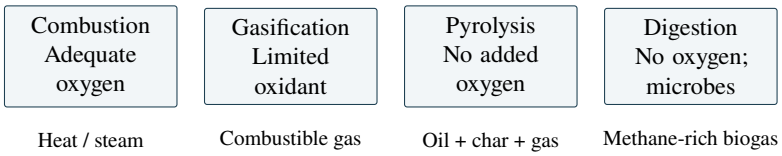
07 | BIOENERGY

Biomass, biofuels and fuel quality

Core idea. Biomass is organic material of biological origin used as an energy resource. It stores chemical energy originally captured mainly by photosynthesis.

Form	Examples	Typical route / product
Solid	Wood, bagasse, husk, straw, pellets	Combustion for heat/steam; gasification for producer gas
Liquid biofuels	Bioethanol, biodiesel, bio-oil	Fermentation, oil transesterification, pyrolysis respectively
Gaseous biofuels	Biogas, upgraded biomethane, producer gas	Digestion; biogas upgrading; partial oxidation/gasification

Conversion routes: do not confuse their oxygen needs



Biofuel types by feedstock generation: first-generation fuels use sugar, starch or edible oil crops; second-generation routes use lignocellulosic material/residues; third-generation commonly refers to algal routes. Definitions beyond these vary, so identify the feedstock and process rather than relying only on a generation number.

Characteristics that matter in a biomass fuel

- **Moisture:** wet fuel consumes heat in evaporation and reduces useful heat per kilogram as fired.
- **Calorific value:** chemical energy released per unit mass; always note wet or dry basis.
- **Ash and ash chemistry:** affect waste handling, deposits, corrosion and clinker formation.
- **Volatile matter and fixed carbon:** influence ignition, flaming and char-burning behaviour.
- **Size and bulk density:** affect feeding, transport, storage, gas flow and conversion uniformity.

Advantages, disadvantages and applications

Advantages: uses residues, supports rural incomes, offers storable fuel, can supply heat and dispatchable electricity. **Disadvantages:** bulky/seasonal supply, collection and drying costs, smoke/particulates, competing land uses and possible deforestation. Climate benefit depends on regrowth, land-use change and the full supply chain; biomass is not automatically carbon-neutral.

Applications: cooking/heating, industrial boilers, combined heat and power, electricity generation, vehicle biofuels and organic-waste treatment. Select the conversion route to match moisture, feedstock and required output. [T10,T11]

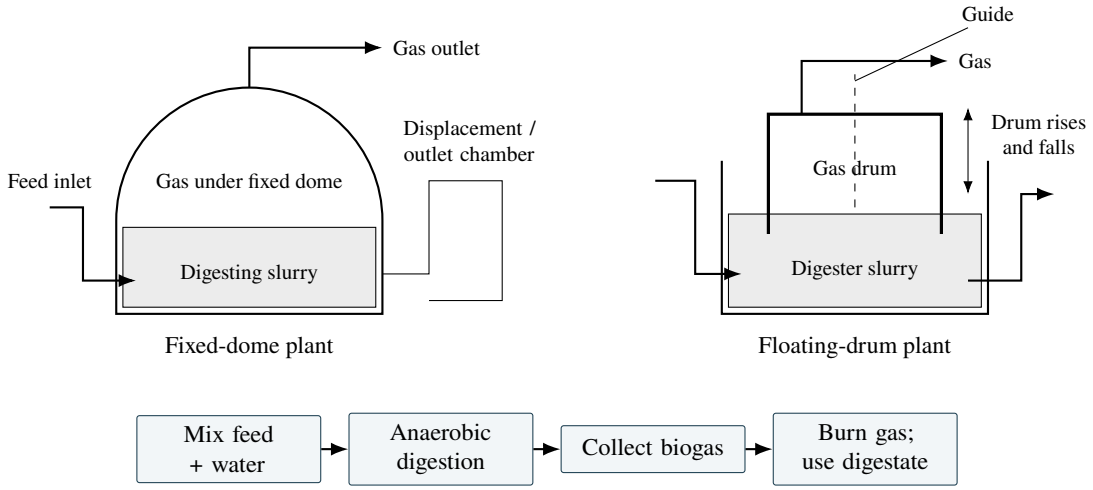
Memory cue. Wet wastes digest; dry residues can burn or gasify. This is a useful first choice, not a rigid rule.

Cover and recall. Name the three fuel forms. Explain why moisture and ash matter, then contrast combustion, gasification and digestion.

08 | Bioenergy systems

Biogas plants: fixed dome and floating drum

Principle. Anaerobic microbes break down wet organic material to produce methane-rich biogas and a nutrient-containing digestate. Typical feedstocks are cattle dung, sewage and food waste.



Biological sequence: hydrolysis breaks complex matter into soluble units; acidogenesis forms organic acids; acetogenesis forms acetate and other intermediates; methanogenesis produces methane. Keep air out and maintain suitable temperature, pH and retention time.

Feature	Fixed dome	Floating drum
Gas holder	Sealed masonry dome	Movable inverted drum, guided vertically
When gas accumulates	Slurry displaced to compensation chamber; pressure increases	Drum rises; pressure approximately constant for a freely moving drum
Main merit	No moving steel holder; low maintenance when well built	Gas volume visible; relatively steady gas pressure
Main limitation	Construction must be gas-tight; varying pressure	Drum cost, corrosion and moving-part maintenance

Applications and benefits: cooking, hot-water heating and electricity in a suitable engine; waste treatment and useful digestate. **Limitations:** regular feed/water supply, slow startup, sensitivity to poor operating conditions, gas leakage and cleaning requirements. Remove moisture and hydrogen sulphide as required before engine use. [T12,T13]

Source wording: “floating dome” in one paper is treated here as the standard floating-drum biogas plant, not a fixed masonry dome.

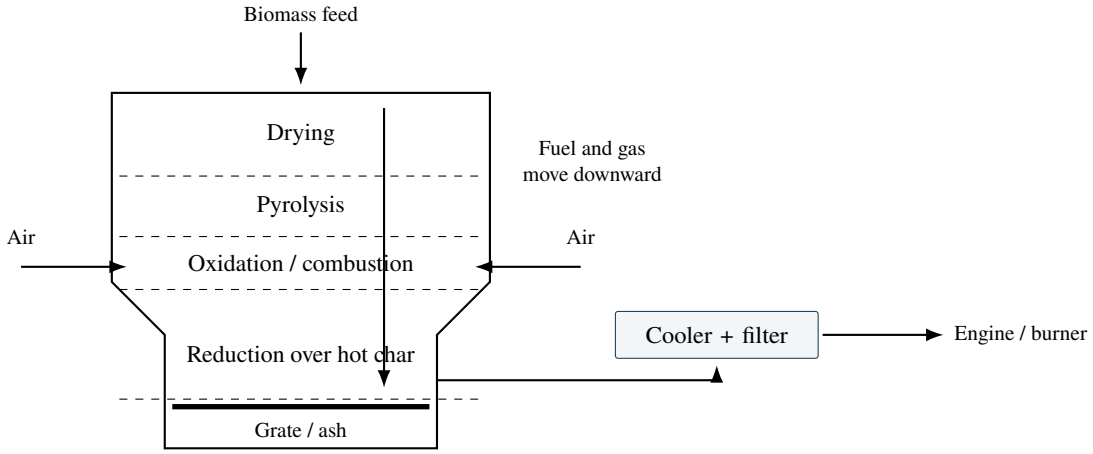
Memory cue. Fixed roof: pressure changes. Floating roof: height changes.

Cover and recall. Draw both gas holders and their slurry outlets. Where does the slurry go when gas accumulates in a fixed-dome plant?

09 | Thermochemical conversion

Downdraft biomass gasifier

Core idea. A gasifier converts solid biomass into combustible producer gas using less oxidant than complete combustion requires. In a downdraft gasifier, fuel and gas move mainly downward together.



Explain the zones in this order

1. **Drying:** incoming biomass loses moisture using heat from the hotter lower regions.
2. **Pyrolysis:** heated biomass breaks into char, gases and condensable vapours/tars.
3. **Oxidation:** air enters near the throat. Part of the fuel/gas burns, releasing heat to sustain drying and gasification.
4. **Reduction:** hot char reacts with carbon dioxide and steam, producing mainly CO and H_2. Gas exits for cooling and cleaning; ash is removed through the grate.

$$C + CO_2 \rightarrow 2CO \qquad C + H_2O \rightarrow CO + H_2$$

Here C is char carbon, CO_2 carbon dioxide, CO carbon monoxide, H_2 hydrogen and H_2O steam. These reduction reactions consume heat. Producer gas also contains nitrogen (with air gasification), some methane and impurities.

Why relatively low tar? Vapours pass through a hot oxidation region and hot char, helping crack/reform tars. This does *not* make the gas tar-free. Cooling and cleaning remain necessary for engine use. [T14,T15]

Merits: comparatively simple small-scale system, useful heat/electricity from suitable residues, lower tar than many updraft arrangements. **Limits:** needs suitable fuel size/moisture, can suffer channeling and clinker, and has pressure drop and gas-cleaning needs. CO-rich gas is toxic, so leakage control matters in practice.

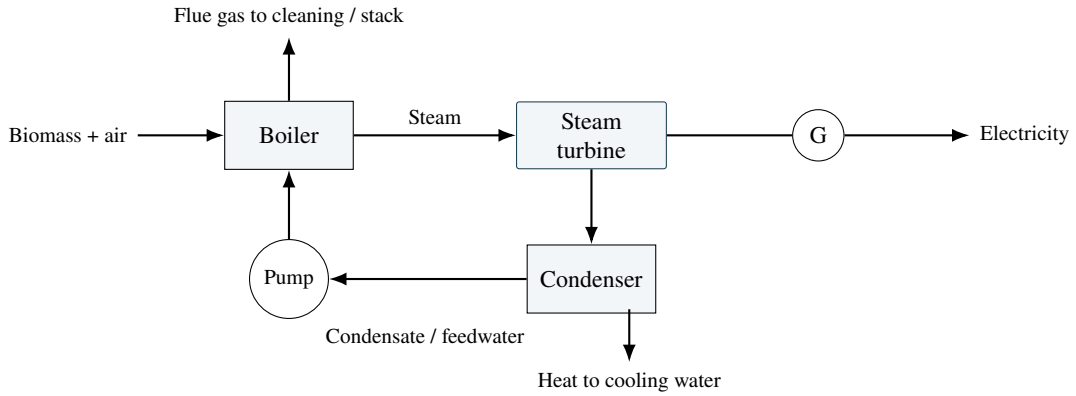
Memory cue. D–P–O–R: Dry, Pyrolyse, Oxidise, Reduce. Oxidation supplies heat; reduction makes fuel gas.

Cover and recall. Draw the throat, air nozzles, four zones, grate and gas outlet. Why must the gas be cleaned before entering an engine?

10 | BIOENERGY POWER SYSTEMS

Biomass power, cogeneration and combined cycle

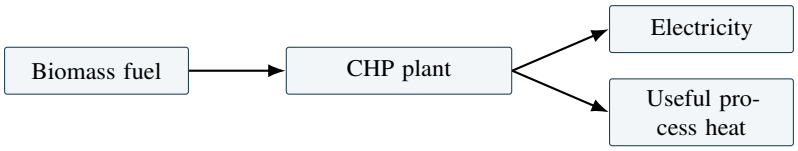
Biomass-fired steam power plant



Working: prepared biomass burns in a furnace; the boiler raises steam; expanding steam turns the turbine and generator; exhaust steam condenses; a feed pump returns water to the boiler. Flue-gas cleaning and ash handling complete the fuel side. The condenser removes waste heat; it is not the steam source.

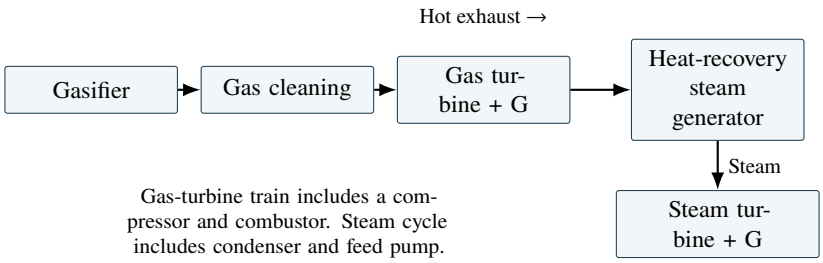
Cogeneration: obtain useful heat as well as power

Cogeneration, or combined heat and power (CHP), delivers electricity/mechanical power *and useful process heat* from one energy input. A sugar mill can burn bagasse, generate power and use steam for process heating. In a back-pressure turbine, exhaust goes to the heat load; an extraction turbine supplies some steam before the remaining flow expands further.



CHP improves total fuel utilisation when a suitable heat demand exists; its benefit falls if heat is unwanted or must travel far.

Biomass integrated gasification combined cycle



Working: cleaned fuel gas burns with compressed air; hot gases expand in the gas turbine. Its exhaust raises steam in the heat-recovery unit, and steam drives a second turbine. **Combined cycle** means these two linked power cycles. It improves heat recovery but adds gas-cleaning, control and capital requirements. CHP describes useful heat plus power; a combined cycle describes two power cycles. [T10]

Memory cue. CHP = two useful outputs. Combined cycle = two power cycles.

Cover and recall. Draw the steam loop. Explain the difference between CHP and combined cycle using a sugar mill and a gasifier-based plant.

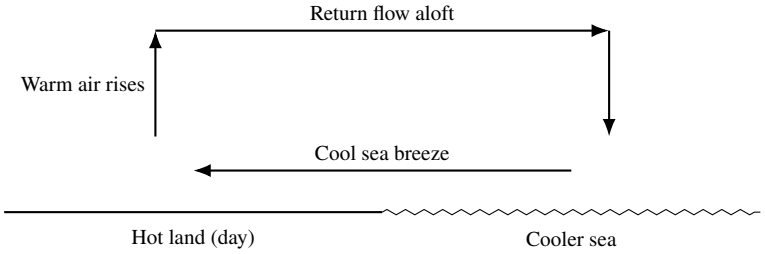
11 | Wind energy

Wind origin and the WECS chain

Core idea. Unequal solar heating creates temperature and pressure differences. Air moves from higher toward lower pressure, with its direction modified by Earth's rotation, terrain and friction. A wind energy conversion system (WECS) extracts part of the air's kinetic energy.

Global and local winds

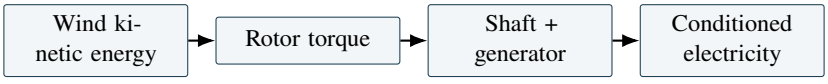
Global winds	Large-scale circulation driven by uneven equator-to-pole heating and Earth's rotation: trade winds, mid-latitude westerlies and polar easterlies.
Local winds	Smaller-scale temperature/terrain effects: land–sea breezes, valley–mountain winds. Monsoon circulation is seasonal and regional in scale.



At night, land commonly cools faster than sea: near-surface flow can reverse to a land breeze. This sketch is a simplified local circulation, not a universal wind forecast.

Basic WECS components and their jobs

Component	Function
Rotor: blades + hub	Aerodynamic forces produce torque.
Drivetrain	Shaft and, where fitted, gearbox transmit torque at a suitable speed; direct-drive systems omit a gearbox.
Generator + converter	Convert mechanical power to electricity and condition its voltage/frequency.
Control and protection	Sensors, controller, pitch/stall control, brake and shutdown functions regulate/safeguard operation.
Support and connection	Nacelle, tower and foundation carry loads; cables, transformer and switchgear connect the output.



Power idea: $\boxed{P_{\text{wind}} = \frac{1}{2}\rho A v^3}$. Here ρ is air density, A swept area, and v undisturbed wind speed. The rotor captures only a fraction; real electrical output also includes losses. This explains why site wind speed matters strongly. [T16]

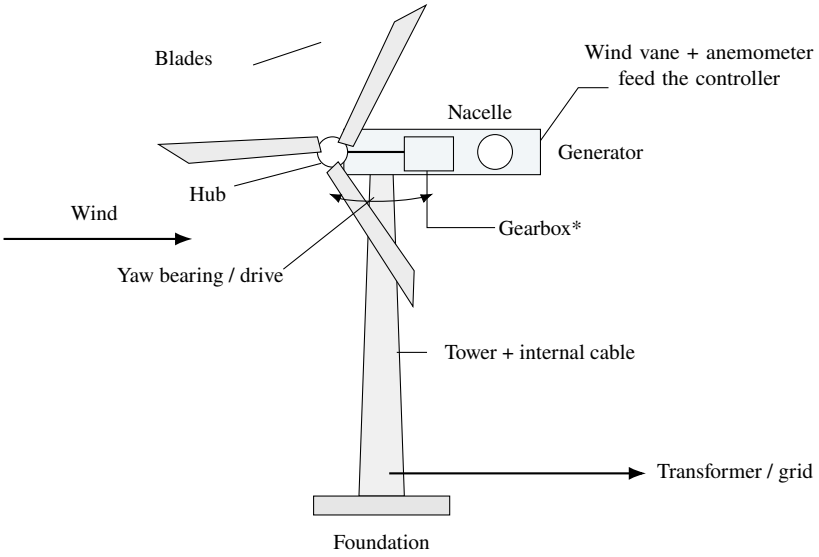
Memory cue. Sun makes pressure differences; rotor turns wind into torque.

Cover and recall. Draw a sea-breeze loop and the WECS chain. Which components convert energy, and which support/control the conversion?

12 | Wind turbine construction

Horizontal-axis wind turbine

Definition. A horizontal-axis wind turbine (HAWT) has its rotor axis approximately horizontal and generally aligned with the wind. The common three-bladed design uses aerodynamic lift.



*Conceptual HAWT and drivetrain. *A direct-drive design omits the gearbox. Rotor and nacelle are shown schematically together to expose the components.*

Working

1. Wind flowing over shaped blades creates lift and drag; the useful tangential force produces rotor torque.
2. The rotor drives a low-speed shaft. A gearbox may increase speed for the generator, or a direct-drive generator is used.
3. The generator produces electrical power; converters and a transformer prepare it for the load or grid.
4. A yaw system turns the nacelle toward the wind. Pitch control changes blade angle to regulate power and aerodynamic loads.
5. The controller starts/stops the machine within its operating range; braking and protective shutdown limit unsafe operation.

Advantages: efficient lift-based conversion, tall towers reach stronger/steadier winds, mature large-machine design. **Limitations:** yaw and control equipment, heavy machinery aloft, difficult access, variable wind and site/environmental constraints. [T16,T17]

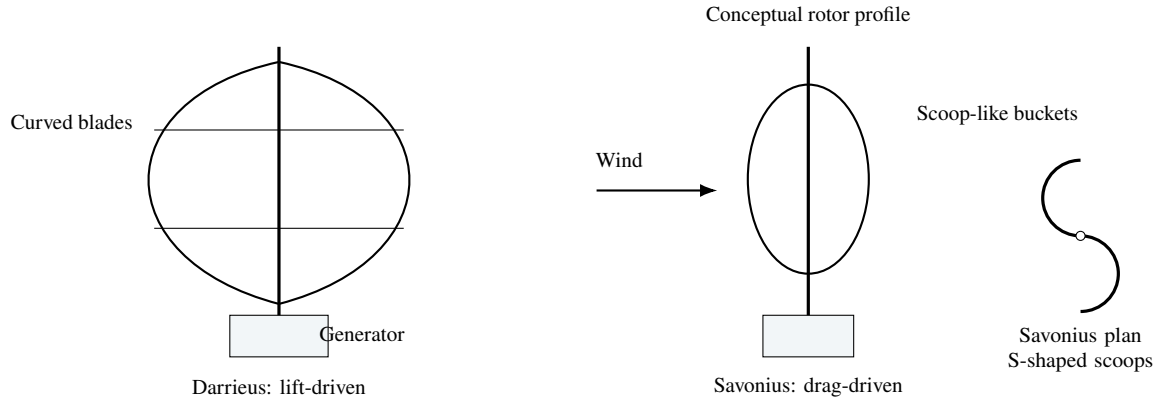
Memory cue. Yaw points the nacelle. Pitch turns a blade. Rotor → shaft → generator → grid.

Cover and recall. Sketch and label at least eight parts. Explain yaw versus pitch, and say why a gearbox is not present in every HAWT.

13 | Wind turbine comparison

Vertical-axis turbines and wind applications

Definition. A vertical-axis wind turbine (VAWT) rotates about a vertical shaft and can accept horizontal wind from different directions without a yaw mechanism.



Working: wind produces unequal aerodynamic forces around the rotating blades/buckets. Their net torque turns a vertical shaft linked to a generator. Darrieus rotors mainly use lift; Savonius rotors mainly use differential drag. Some Darrieus machines need starting assistance; Savonius machines usually start readily but have lower efficiency.

Feature	HAWT	VAWT
Rotor axis	Horizontal	Vertical
Wind alignment	Usually requires yaw	No yaw for horizontal direction changes
Machinery position	Usually in nacelle aloft	Can place some equipment near ground
Main trade-off	High-performance mature designs; tall support	Easier direction acceptance; cyclic loads and lower near-ground wind may be concerns

Wind energy: benefits, limitations, uses

Benefits: no fuel purchase during operation, low operational emissions, short modular additions, land beneath some farms remains usable. **Limitations:** variable output, site and transmission needs, visual/noise effects, wildlife concerns and maintenance access. Careful siting and controls reduce impacts; they do not eliminate all impacts.

Applications: grid electricity, battery charging, water pumping and remote/hybrid supplies. A wind pump may drive a mechanical pump directly; an electrical system uses a generator and motor-driven pump. [T16]

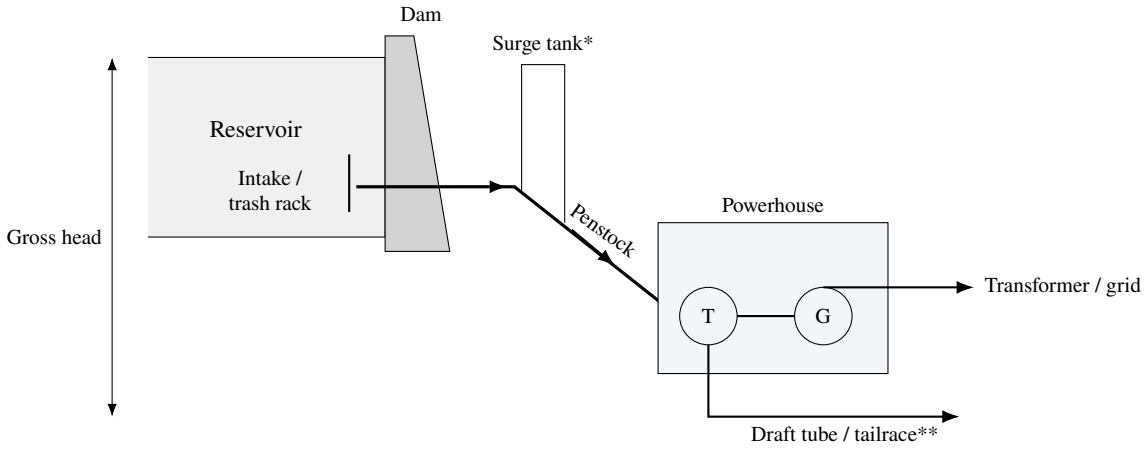
Memory cue. Darrieus = lift; Savonius = scoops and drag.

Cover and recall. Draw both rotors, compare HAWT/VAWT in four rows, and list two wind applications other than supplying the grid.

14 | Hydropower

Hydroelectric power plant

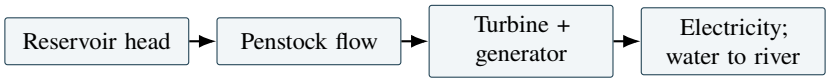
Principle. Water falling through a usable head converts gravitational energy into turbine shaft work and then electricity. Power depends on both discharge and *net* head.



*T = turbine; G = generator. *Surge tanks are used where required to limit pressure transients. **A draft tube applies to reaction turbines; a Pelton runner discharges freely.*

Construction and working

1. A dam/reservoir stores water and provides head; a spillway passes excess water safely.
2. An intake with trash rack and gate admits water while excluding large debris and allowing isolation.
3. The penstock carries pressurised water to the turbine. A surge tank, where provided, moderates rapid flow/pressure changes.
4. Water transfers energy to the runner; the shaft drives a generator.
5. A reaction turbine discharges through a draft tube to the tailrace. A transformer raises electrical voltage for transmission.



$$P = \rho g Q H_{\text{net}} \eta$$

Here P is output power, ρ water density, g gravitational acceleration, Q discharge, H_{net} head after hydraulic losses, and η overall efficiency. [T18]

Merits: high conversion efficiency, low operational emissions, long life, quick response; reservoir plants can store water/energy. **Limitations:** high initial civil cost, site dependence, hydrological variation, siltation, fish/river impacts and possible displacement. Run-of-river plants have less storage and less control over seasonal supply.

Memory cue. Store → guide → spin → generate → return.

Cover and recall. Draw the complete water path. Distinguish penstock, surge tank and draft tube, and explain gross versus net head.

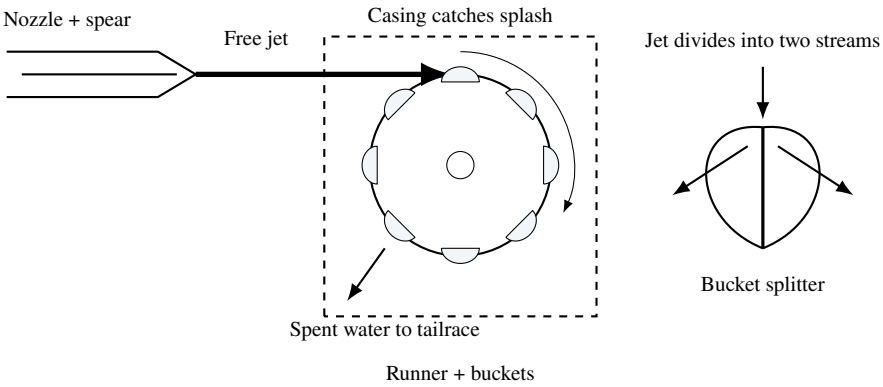
15 | Impulse turbine

Hydraulic turbine classification and Pelton wheel

Classify hydraulic turbines on four bases

Basis	Classes / examples
Energy at runner	Impulse: Pelton. Reaction: Francis, Kaplan.
Flow through runner	Tangential: Pelton; mixed (radial inlet, axial outlet): Francis; axial: Kaplan/propeller.
Suitable head and flow	Relatively high head/low flow: Pelton; medium range: Francis; low head/high flow: Kaplan. Ranges overlap.
Shaft orientation	Horizontal or vertical, depending on design and installation.

Impulse versus reaction. An impulse turbine converts head into a fast free jet before the runner; the runner operates approximately at atmospheric pressure. A reaction turbine extracts energy while pressure changes within filled runner passages and needs a suitable enclosed flow path.



Pelton working and important parts

1. The nozzle converts pressure head into jet velocity; a spear/needle regulates flow.
2. The jet strikes the bucket splitter tangentially and divides into two streams.
3. Curved buckets turn the water through a large angle; the change in momentum produces torque.
4. The runner drives the generator, and spent water falls freely to the tailrace.

Parts to label: penstock/nozzle, spear, jet, splitter, double-cup buckets, runner, shaft and casing. The casing prevents splashing; it is not a pressure casing. **No draft tube** is used for energy recovery from a free-discharge Pelton runner. [T19,T20]

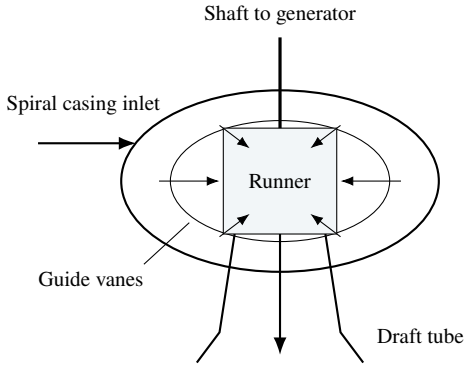
Memory cue. Pelton: a jet pushes split buckets.

Cover and recall. Classify Pelton, Francis and Kaplan by action and flow. Why does a Pelton turbine have a nozzle but no draft tube?

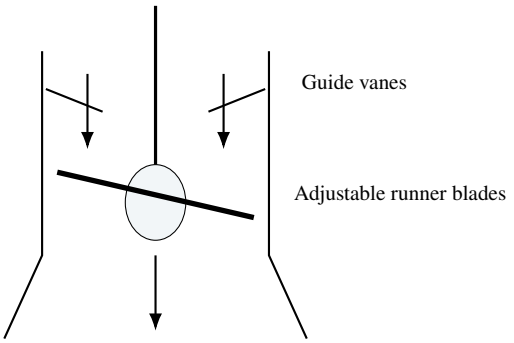
16 | Reaction turbines

Francis and Kaplan turbines

Common principle. Both are reaction turbines: water completely fills their passages, and pressure and velocity change as energy is transferred to the runner. Guide vanes control admission; a draft tube recovers part of the exit kinetic energy and connects to the tailrace.



Francis: radial inward → axial out



Kaplan: axial in → axial out

Simplified flow sections; arrows show flow through the runner rather than an exact 3D blade shape.

Working sequences

Francis: penstock → spiral casing distributes water around the runner → stay/guide vanes regulate and direct flow → water enters the runner mainly radially inward and leaves axially → draft tube discharges to tailrace.

Kaplan: casing and guide vanes admit water → flow reaches the propeller-like runner approximately axially → adjustable runner blades produce torque → axial discharge passes through the draft tube. Coordinating guide-vane opening and blade pitch maintains efficiency over changing load/flow.

Feature	Francis	Kaplan
Runner flow	Mixed: radial in, axial out	Axial
Runner blade control	Usually fixed runner blades; adjustable guide vanes	Adjustable runner blades and guide vanes
Typical suitability	Intermediate head/discharge	Lower head, larger discharge
Common concern	Cavitation and off-design losses	Cavitation, pitch mechanism and flow control

Draft tube purpose: its expanding passage slows discharge, recovering part of the velocity head as pressure. Correct installation and pressure control help avoid cavitation. Exact head limits depend on the turbine design, not a universal cut-off. [T19,T20,T21]

Memory cue. Francis turns the flow; Kaplan keeps it axial.

Cover and recall. Draw both runner flow directions. Which blades are adjustable? Explain the draft tube in one sentence without calling it a nozzle.

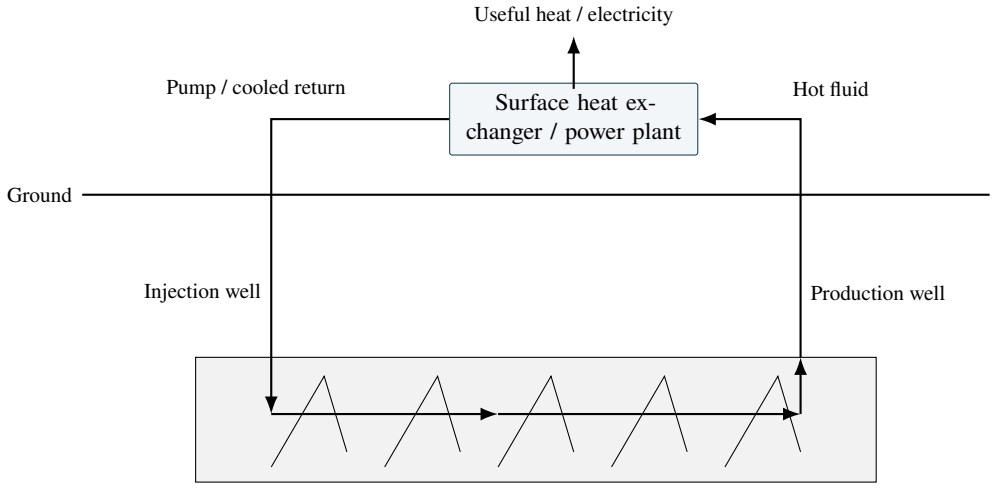
17 | Geothermal energy

Geothermal resources and hot dry rock

Core idea. Geothermal energy is heat from within the Earth. A conventional hydrothermal resource needs *heat, fluid and permeability* so hot fluid can circulate and be produced.

Resource	What is present / how used
Vapour-dominated	Mainly steam in productive zones; suitable steam may drive a dry-steam plant.
Liquid-dominated	Hot water/brine; flash steam at suitable conditions or transfer heat to a binary cycle.
Hot dry rock / EGS	Hot rock with inadequate natural fluid or permeability; engineered circulation extracts heat.
Geopressured	Deep hot fluids under high pressure, sometimes with dissolved gas; technically demanding.
Magma / very hot rock	Large heat potential, difficult materials/drilling conditions; not a routine conventional resource.

Hot dry rock: enhanced geothermal system (EGS)



Hot fractured rock: water circulates through connected pathways

Working: drill injection and production wells; establish or improve connected flow paths; pump fluid into the hot rock; fluid absorbs heat and returns through a production well; transfer heat at the surface and reinject the cooled fluid. The rock supplies heat, while the circulating fluid transports it. EGS is distinct from a sealed underground pipe loop. [T22]

Benefits, applications and limits

Benefits: reliable site-based heat supply, small land footprint relative to some alternatives, no purchased combustion fuel for the heat source. **Applications:** electricity at suitable temperatures, district/process heating, greenhouses, aquaculture and bathing; ground-source heat pumps use shallow ground as a heat source/sink and need electricity.

Limits: high exploration/drilling cost, geographic constraints, reservoir cooling/depletion if poorly managed, scaling/corrosion, dissolved gases and induced-seismicity concerns for some developments. Reinjection supports fluid management but does not guarantee unlimited reservoir output. [T23]

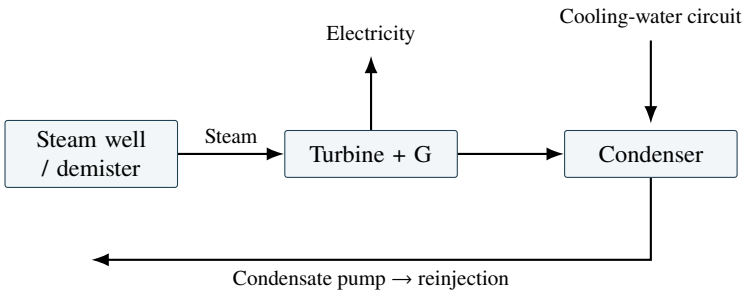
Memory cue. Heat + fluid + pathways. EGS supplies the missing circulation conditions.

Cover and recall. Name four geothermal resource types. Draw the two-well HDR system and explain why reinjection is needed.

18 | Geothermal power cycles

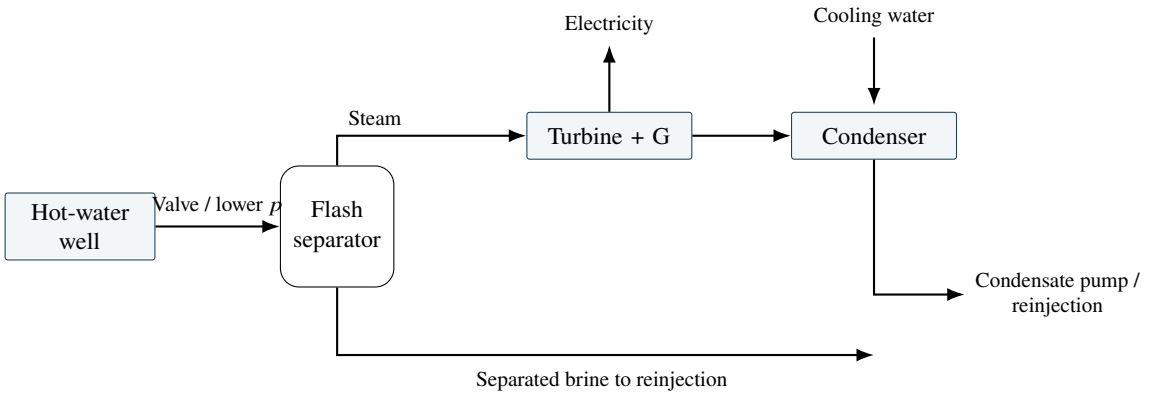
Dry-steam and single-flash geothermal plants

Dry-steam plant: natural steam goes to the turbine



Working: produce naturally occurring steam; remove entrained droplets/solids; expand steam through a turbine-generator; condense the exhaust; manage non-condensable gases and reinject condensate as designed. A suitable vapour-dominated reservoir is required. There is no fossil-fired boiler creating the steam.

Wet/liquid-dominated resource: single-flash plant



1. Bring pressurised hot geothermal water to the surface.
2. Reduce its pressure. At the lower pressure, a fraction of the hot water *flashes* into steam; this is not a sudden conversion of all water into steam.
3. A separator sends steam to the turbine and residual liquid to brine handling/reinjection.
4. Condense turbine exhaust, remove gases as needed, and return fluids to the reservoir through the reinjection system.

Terminology: a "wet-steam plant" question usually asks how a liquid/two-phase geothermal supply is separated before turbine admission. A flash separator supplies relatively dry steam; sending liquid droplets through a steam turbine damages performance and blades.

Comparison: dry steam uses reservoir steam directly; single flash obtains steam from hot liquid by reducing pressure. Both need a suitable resource, turbine, generator, condenser/cooling system and fluid management. [T23,T24]

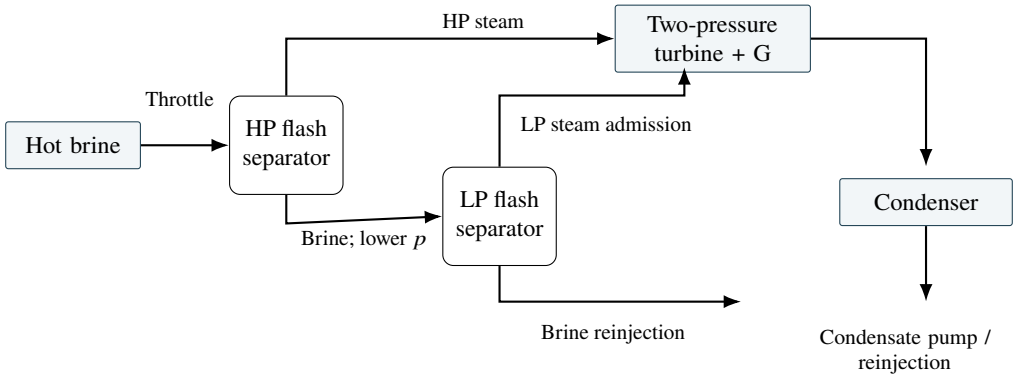
Memory cue. Dry = steam already present. Flash = pressure drops, some water boils.

Cover and recall. Redraw both paths. What enters the separator, what leaves its top, and what leaves its bottom?

19 | Geothermal cycle comparison

Double flash and binary geothermal cycles

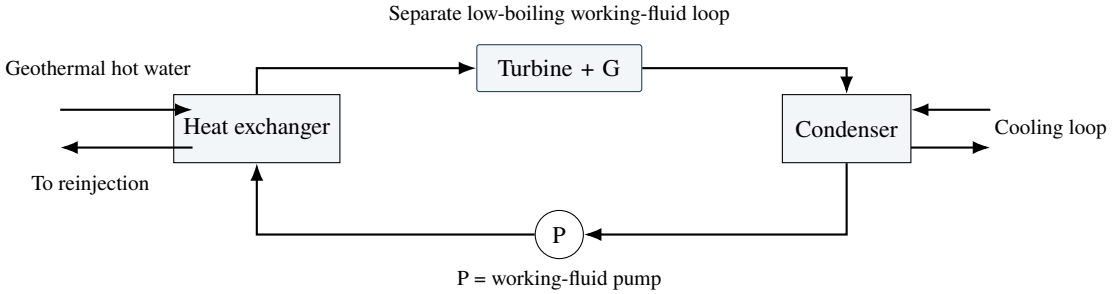
Double flash: recover steam at two pressures



Working: first flash at higher pressure separates HP steam. Remaining liquid is throttled to a lower pressure, creating more steam. LP steam enters a suitable lower-pressure turbine stage. Exhaust steam condenses; residual brine and condensate are managed and reinjected. HP means high pressure; LP means low pressure.

Why use it? More steam and energy can be recovered from the same hot-water stream than with one flash stage, but extra vessels, piping and controls increase cost and complexity. More steam recovery does not mean all reservoir heat becomes electricity.

Binary cycle: transfer heat to a separate fluid



Key difference: geothermal fluid heats a secondary working fluid in a heat exchanger and returns to reinjection; it does not pass through the turbine. The secondary fluid evaporates, expands, condenses and is pumped around a closed loop. Binary plants can use resources cooler than those typically used for flash generation, depending on design. [T23,T24,T25]

Dry steam	Single flash	Double flash	Binary
Natural steam	One separation pressure	Two separation pressures	Heat exchanger; secondary fluid

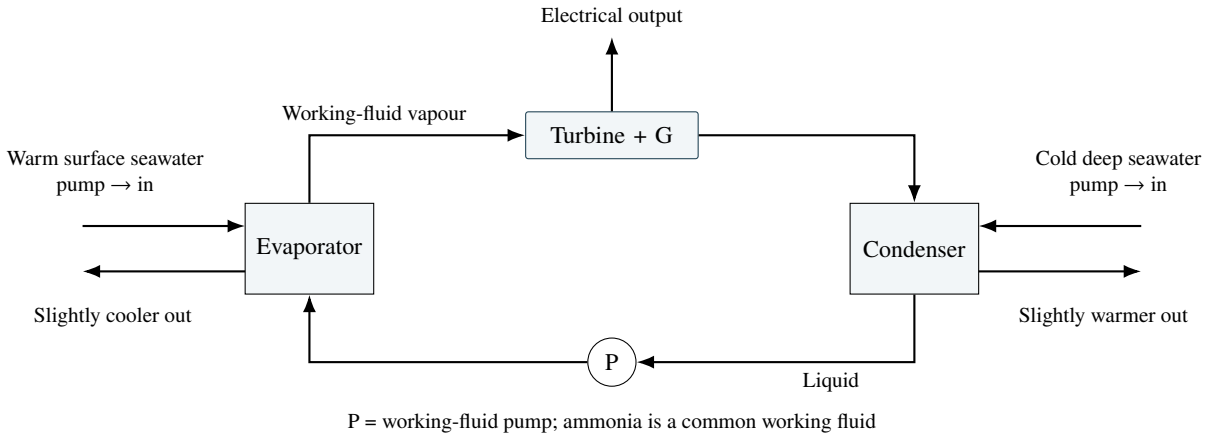
Memory cue. Count separators for flash; look for a second fluid for binary.

Cover and recall. Draw the two steam admissions of double flash. Why must the binary working fluid remain separate from geothermal brine?

20 | Ocean energy

Closed-cycle ocean thermal energy conversion

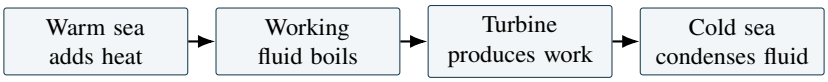
Principle. OTEC uses warm surface seawater as a heat source and cold deep seawater as a heat sink. A temperature difference of roughly 20°C or more is commonly sought for practical designs; useful sites are mainly tropical.



Three separate circuits: warm seawater, cold seawater, and the recirculating working fluid. Seawater does not enter this turbine.

Working sequence

1. Warm seawater transfers heat in the evaporator to a low-boiling working fluid such as ammonia.
2. The working fluid vaporises and expands through a turbine connected to a generator.
3. Cold deep seawater removes rejected heat in the condenser and turns the vapour back into liquid.
4. A pump returns the working liquid to the evaporator. Separate seawater pumps maintain the warm and cold flows.



Benefits and applications

Benefits: the ocean stores solar heat; tropical temperature differences can be comparatively steady; generation uses no combustion fuel. Potential integrated uses include electricity, cooling with deep seawater, and aquaculture. Freshwater production is associated with open/hybrid arrangements or added desalination, not an automatic product of a simple closed ammonia loop.

Main engineering limits: small temperature difference gives low thermal efficiency; large seawater flows, deep-water piping, corrosion, fouling and storm-resistant offshore works are demanding. Pump power must be subtracted from turbine-generator output to obtain net output. Discharge and intake design must consider marine ecosystems. [T26,T27]

Energy logic: warm source → engine → cold sink. Some heat must be rejected; the plant cannot convert all absorbed heat into work.

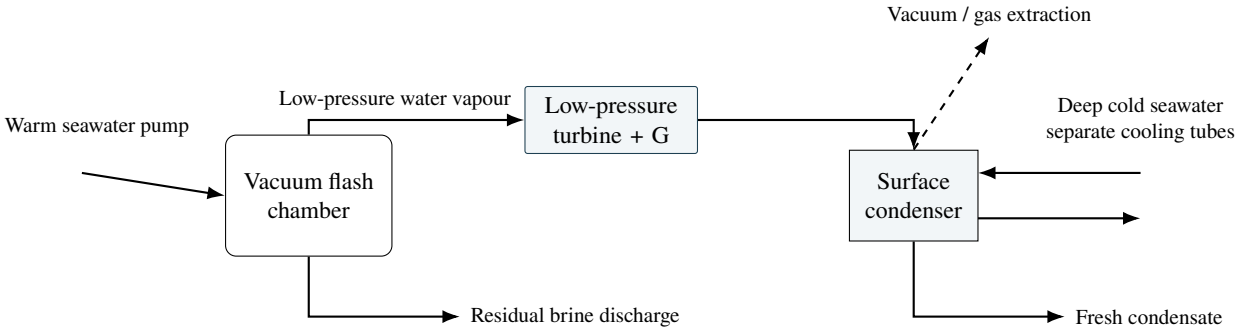
Memory cue. Closed = a circulating working fluid. Warm boils it, cold condenses it.

Cover and recall. Draw all three circuits separately. Why is a condenser essential, and why can gross turbine output overstate useful OTEC power?

21 | Ocean energy comparison

Open-cycle OTEC and ocean-energy limits

Core idea. In open-cycle OTEC, warm *seawater itself* supplies the water vapour that drives a low-pressure turbine. The plant operates under a strong vacuum so warm water can flash at a low temperature.



Working

1. Pump warm surface seawater into a low-pressure chamber; a small fraction flashes into water vapour.
2. Salt remains mainly in the residual liquid. Route the vapour through the turbine-generator.
3. Condense the exhaust using cold deep seawater. A *surface condenser* keeps coolant separate and permits fresh-water collection.
4. Remove non-condensable gases to maintain the vacuum; discharge brine and used seawater appropriately.

If steam directly mixes with seawater coolant in a direct-contact condenser, its condensate is not separately recovered as fresh water. State the condenser arrangement when claiming desalination. [T26,T27]

Feature	Open OTEC	Closed OTEC
Turbine fluid	Water vapour from seawater	Separate low-boiling fluid
Evaporation	Seawater flashes under vacuum	Warm sea heats secondary fluid
Fresh water	Possible with separate surface condensation	Needs additional/hybrid desalination
Common difficulty	Vacuum, low-density vapour and large equipment	Heat exchangers and working-fluid management

Ocean energy: forms, merits and limitations

Forms: OTEC uses a temperature difference; waves use surface motion; tides use periodic sea-level/current changes. All are site-dependent marine resources, with different conversion devices.

Merits: large natural resource, low operational fuel use, useful coastal/island applications; tides are predictable and OTEC can be comparatively steady at suitable sites. **Limits:** difficult access, mooring/civil cost, saltwater corrosion, biofouling, storms, grid distance and ecological/navigation conflicts. Wave variability is different from OTEC's low-efficiency problem.

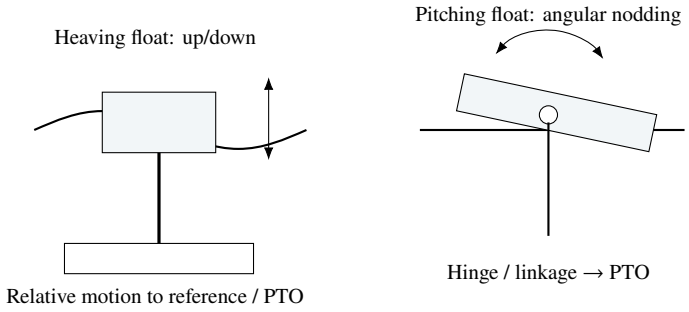
Memory cue. Open = ocean water becomes steam. Closed = another fluid circulates.

Cover and recall. Draw open OTEC and show where salt stays. Under what condenser condition is fresh water recovered?

22 | WAVE ENERGY

Wave devices: heaving, pitching and OWC

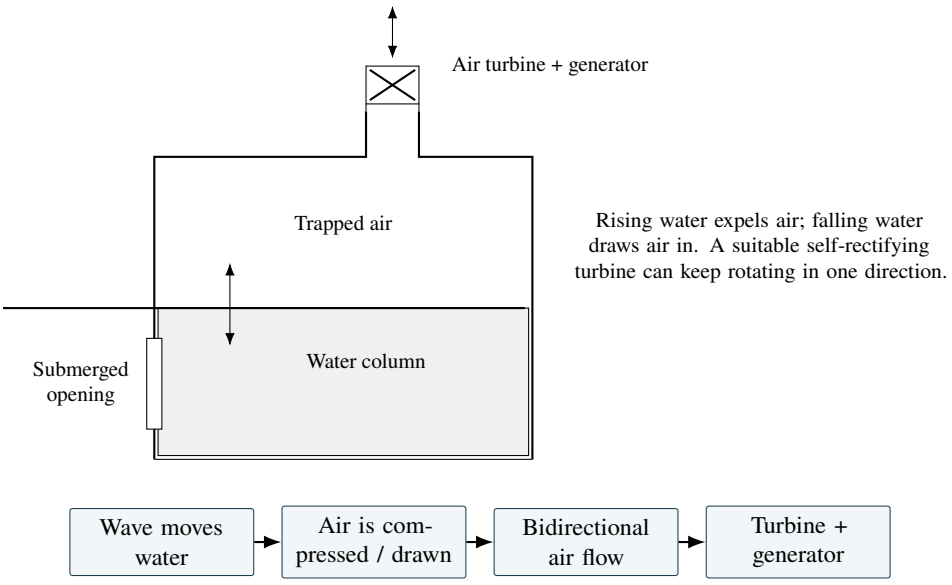
Core idea. Wave devices convert oscillatory motion into mechanical or pneumatic power, then electricity. The power take-off (PTO) is the mechanism that performs this conversion.



Heaving device: a buoy moves vertically with waves relative to a seabed-fixed or more stable reference. A hydraulic system, mechanical linkage or linear generator converts relative motion to electrical output. The body needs relative motion against something; merely drifting with the water is not a complete conversion system.

Pitching device: a floating body rocks about a horizontal transverse axis. A hinge or linkage drives a hydraulic/rotary PTO. The useful motion is *angular*, not vertical translation.

Oscillating water column (OWC)



Uses and limits: coastal/island electricity and possible power for marine loads/desalination systems; output varies with sea state. Storm survival, moorings, corrosion, maintenance and power smoothing are important challenges. [T27]

Memory cue. Heave = lift; pitch = nod; OWC = breathe.

Cover and recall. Draw three distinct motions. In an OWC, does air flow only one way? How can turbine rotation still keep one direction?

23 | ENGINEERING ECONOMICS

Project costs and cash-flow diagrams

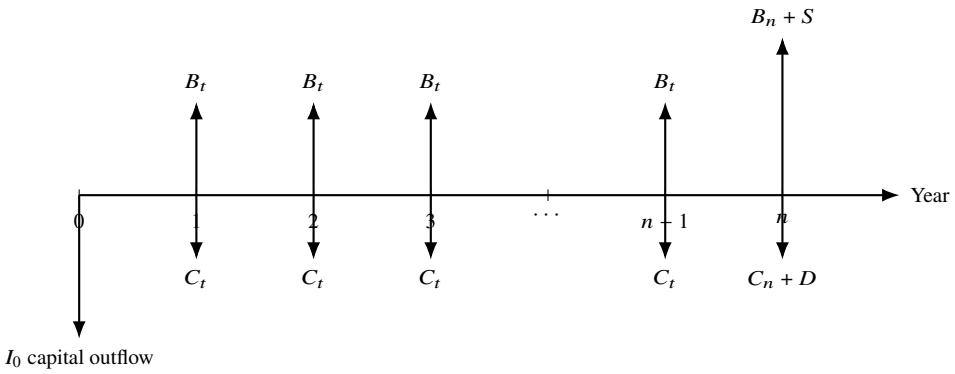
Core idea. Compare costs and benefits at their actual times. A cash-flow diagram records *when* money is spent or received, not just its total.

Initial, annual and end-of-life costs

Cost category	Typical project items
Initial / capital cost	Land, civil works, plant equipment, installation, engineering, approvals, commissioning and initial connection.
Annual operating cost	Fuel/feedstock where required, labour, routine maintenance, insurance, administration and recurring charges.
Periodic replacements	Major overhauls, inverter/battery/part replacements at their expected years; not necessarily equal annual amounts.
End-of-life cash flow	Salvage value or sale proceeds, less dismantling, disposal and site-restoration expenditure.

Other useful distinctions: fixed costs do not vary strongly with short-term output; variable costs change with production. Direct costs can be assigned to the project/activity; indirect costs support it more broadly. A sunk cost is already incurred; an opportunity cost is the value of the best forgone alternative. State which costs enter the particular analysis.

Cash-flow sketch: take the investor's viewpoint



Upward arrows: receipts; downward arrows: payments. I_0 initial investment, B_t benefit/revenue in year t, C_t operating cost, S salvage receipt, D decommissioning cost, n final year.

1. Draw a time line and mark year 0 and all payment/receipt years.
2. Place capital expenditure at its specified time; show annual benefits and costs separately or as clearly labelled net cash flows.
3. Add replacements, salvage and closure costs at their actual dates.
4. State the sign convention and the analysis viewpoint. Do not count the same cost twice.

Cash flow versus accounting profit: depreciation reduces accounting profit but is not itself a cash payment; it may affect tax. Debt interest/principal depend on whether the analysis is project or equity based. Use assumptions consistently instead of mixing viewpoints. [T28,T29]

Memory cue. Cost has a name, a sign and a date.

Cover and recall. Draw a five-year project with capital cost, annual savings, one replacement and final salvage. Which arrows point down?

24 | Engineering economics

Financial appraisal: value, rate and recovery

Notation. I_0 = initial investment; F_t = net cash inflow at end of year t; i = discount rate per year; n = project life. Include final salvage/closure in F_n where applicable. Use consistent currency, time period and real/nominal assumptions.

Net present value (NPV)

$$\boxed{\text{NPV} = -I_0 + \sum_{t=1}^{n} \frac{F_t}{(1+i)^t}}$$

Discount future net cash flows to year 0, then subtract investment. Positive NPV means value is added at the selected discount rate; zero means the required return is just met. For mutually exclusive feasible options, compare at consistent lives/assumptions and the appropriate decision basis.

Merit: recognises time value and all included cash flows. **Limitation:** sensitive to the discount rate and uncertain future estimates. A large project can have a larger NPV even with a lower percentage return.

Simple payback period (PBP)

$$\boxed{\text{PBP} = \frac{I_0}{F} \quad \text{for constant annual net cash inflow } F > 0}$$

For unequal inflows, cumulate cash flows until investment is recovered; interpolate within the recovery year only if uniform flow during that year is assumed. **Merit:** simple measure of recovery/liquidity. **Limits:** ignores the time value of money and cash flows after payback; it is not a complete profitability measure. Discounted payback uses discounted flows but still ignores later benefits.

Return on investment (ROI): state the convention

$$\boxed{\text{ROI} = \frac{\text{average annual net accounting profit}}{\text{initial investment}} \times 100\%}$$

This is one common simple/accounting convention; some courses use average investment or a total-period return. State numerator, denominator and period. **Merit:** easy percentage comparison. **Limits:** convention-dependent, does not directly account for timing, and can confuse profit with cash flow. It is not the same as IRR.

Internal rate of return (IRR)

$$\boxed{0 = -I_0 + \sum_{t=1}^{n} \frac{F_t}{(1+r)^t}}$$

IRR is the rate r that makes NPV zero. For a conventional investment with a unique IRR, compare it with the required return. **Merit:** expresses return as a rate with time value included. **Limits:** unusual cash-flow sign changes can cause multiple/no IRRs; rankings can differ from NPV for different project scales/timings. [T28,T29]

Question	When is money recovered?	What annual percentage?	What value today?
Quick answer	Payback	ROI / IRR (define which)	NPV

Memory cue. Payback = when. ROI = stated accounting rate. NPV = value. IRR = zero-NPV rate.

Cover and recall. Give one advantage and one limitation of each method. Why can the shortest payback still be the poorer long-term investment?

Revision sheet

Closed-book revision and diagram drill

A compact revision sequence

Work through one family at a time initially. Later, place confusing pairs side by side and explain *why* their flow paths differ. The order below is a practical revision plan, not a prediction of examination questions.

Session	Retrieve first, then check	Minimum drawing set
1: Solar	Angles, collector types, thermal applications, PV	Angle sketch; FPC; water heater; dryer; PV junction
2: Biomass	Fuel forms, quality, conversion routes, plant types	Two biogas plants; gasifier zones; steam/combined cycle
3: Wind + hydro	Wind origin, WECS parts, turbine selection	HAWT; two VAWTs; hydro layout; three hydraulic turbines
4: Geothermal	Resources and direct/flash/binary distinctions	Two-well HDR; dry; single flash; double flash; binary
5: Ocean	Heat-source/sink logic and wave motions	Open/closed OTEC; heave; pitch; OWC
6: General + economics	Need, alternatives, costs and appraisal	Classification tree; cash flow; four appraisal formulas

Six common confusions: answer without looking

1. **Yaw or pitch?** Which turns the nacelle, and which changes blade angle?
2. **Flash or binary?** Which sends geothermal-derived steam through a turbine, and which sends a secondary working fluid?
3. **Open or closed OTEC?** Which uses seawater-derived steam? When is fresh-water collection possible?
4. **Cogeneration or combined cycle?** Does the name describe useful heat plus power, or two linked power cycles?
5. **Impulse or reaction?** Which uses a free jet; which has a pressure change through filled runner passages?
6. **Payback or NPV?** Which ignores post-recovery cash flows; which discounts all included cash flows?

A two-minute drawing test

Choose a random system. Draw its boundary and energy input first. Add major parts, connect the working-fluid path, add output and rejected heat/return path, then label control or auxiliary equipment. Reopen the page and mark missing *connections*, not only missing nouns.

Self-check (practice rubric, not an official marking scheme): 1 point for correct parts, 1 for correct arrows, 1 for a correct working sequence, 1 for explaining two parts' functions. A weak score means the next session starts with that drawing.

Short answer key. (1) Yaw: nacelle; pitch: blade. (2) Flash: steam from hot geothermal liquid; binary: separate fluid. (3) Open; freshwater with separation from coolant. (4) CHP: useful heat and power; combined cycle: two cycles. (5) Impulse: free jet; reaction: pressure drop in runner. (6) Simple payback ignores later flows; NPV discounts included life-cycle flows.

Memory cue. If you cannot explain an arrow, do not memorise it blindly. Ask what causes the flow or conversion.

Cover and recall. Pick three weak topics now. Draw them today and retrieve them again after a gap, checking every attempt.

Audit trail | All 99 theory subparts

Coverage index for the seven papers

This index maps every theory subpart to a topic sheet. Mixed questions can point to two sheets. Numericals are excluded. Paper letters follow the PDF order; source page 12 is blank. Question wording is consolidated into topic names rather than reproduced verbatim.

A | 07 February 2023 | PDF pp. 1–2 | 14 theory subparts

1(a) resources [p.3]; 1(b) OTEC benefits [pp.22–23]; 1(c) costs [p.25]; 1(d) biomass merits [p.9]; 2(a) solar angles [p.5]; 2(b) cash flow [p.25]; 3(a) biomass steam plant [p.12]; 4(a) hydro [p.16]; 4(b) dry steam [p.20]; 5(b) alternatives [pp.3–4]; 6(a) dryer [p.7]; 6(b) biofuels [p.9]; 7(a) HAWT [p.14]; 7(b) pitching [p.24].

B | 10 December 2024 | PDF pp. 3–5 | 13 theory subparts

1(a) classification [p.3]; 1(b) HDR [p.19]; 1(c) costs [p.25]; 1(d) biofuels [p.9]; 3(a) fixed dome + biomass [pp.9–10]; 3(b) angles [p.5]; 4(a) hydraulic classification + VAWT [pp.17,15]; 4(b) heave/pitch + wave uses [p.24]; 5(b) nuclear/solar alternatives [p.4]; 6(a) water heater [p.7]; 6(b) cogeneration [p.12]; 7(a) WECS [p.13]; 7(b) open OTEC + ocean limits [p.23].

C | 16 February 2024 | PDF pp. 6–8 | 16 theory subparts

1(a) energy crisis [p.3]; 1(b) dryer [p.7]; 1(c) biomass forms [p.9]; 1(d) ROI [p.26]; 2(a) biomass quality [p.9]; 2(b) Kaplan [p.18]; 2(c) renewables merits/limits [pp.3–4]; 3(b) OWC [p.24]; 3(c) payback [p.26]; 4(c) India need [p.3]; 5(a) dry/wet geothermal [p.20]; 6(a) collectors/FPC [p.6]; 6(b) open OTEC [p.23]; 6(c) wind uses/merits [p.15]; 7(a) angles [p.5]; 7(b) WECS [p.13].

D | 25 February 2025 | PDF pp. 9–11 | 13 theory subparts

1(a) current status [p.4]; 1(b) OTEC benefits [p.22]; 1(c) financial methods [p.26]; 1(d) biomass quality [p.9]; 3(a) floating "dome" + uses [pp.10,9]; 3(b) FPC/classification [p.6]; 4(a) Francis + wind merits [pp.18,15]; 4(b) double flash + ocean uses [pp.21,23]; 5(b) tar sands/oil shale [p.4]; 6(a) solar applications [p.7]; 6(b) biomass combined cycle [p.12]; 7(a) HAWT [p.14]; 7(b) closed OTEC/ocean merits [pp.22–23].

E | 04 December 2023 | PDF pp. 13–15 | 16 theory subparts

1(a) solar alternative [p.4]; 1(b) water heater [p.7]; 1(c) downdraft [p.11]; 1(d) NPV [p.26]; 2(a) combined cycle [p.12]; 2(b) hydro [p.16]; 2(c) renewable/non-renewable [p.3]; 3(b) pitching [p.24]; 3(c) costs [p.25]; 4(c) global scenario [p.4]; 5(b) single/double flash [pp.20–21]; 6(a) PV [p.8]; 6(b) closed OTEC [p.22]; 6(c) VAWT [p.15]; 7(a) geometry [p.5]; 7(b) hydraulic classification + HAWT [pp.17,14].

F | 10 December 2025 | PDF pp. 16–18 | 13 theory subparts

1(a) alternatives [p.4]; 1(b) OTEC benefits [p.22]; 1(c) costs [p.25]; 1(d) biomass forms [p.9]; 3(b) angles [p.5]; 4(a) VAWT/wind merits [p.15]; 4(b) geo resources [p.19]; 5(a) hydro + winds [pp.16,13]; 5(b) alternatives merits [p.4]; 6(a) FPC [p.6]; 6(b) floating drum/uses [pp.10,9]; 7(a) hydraulic classification + WECS [pp.17,13]; 7(b) closed OTEC/ocean limits [pp.22–23].

G | 01 December 2022 | PDF pp. 19–20 | 14 theory subparts

1(a) current scenario [p.4]; 1(b) geo benefits [p.19]; 1(c) costs [p.25]; 1(d) biomass forms [p.9]; 2(a) PV [p.8]; 2(b) cash flow [p.25]; 3(a) combined cycle [p.12]; 4(a) Pelton [p.17]; 4(b) single flash [p.20]; 5(b) renewable need/non-renewable demerits [p.3]; 6(a) CPC [p.6]; 6(b) downdraft [p.11]; 7(a) VAWT [p.15]; 7(b) heaving [p.24].

References | Memory, data and engineering

Sources and further reading

The supplied scanned question-paper compilation determines the coverage and repeat counts. The sources below support terminology, operating principles and study methods. Explanations and vector drawings are original study summaries, not copied textbook figures. Web sources checked 5 November 2026.

Learning and memory research

M1 Roediger, H. L., and Karpicke, J. D. (2006). Test-enhanced learning: Taking memory tests improves long-term retention. *Psychological Science*, 17(3), 249–255. doi:10.1111/j.1467-9280.2006.01693.x.

M2 Cepeda, N. J., Vul, E., Rohrer, D., Wixted, J. T., and Pashler, H. (2008). Spacing effects in learning: A temporal ridgeline of optimal retention. *Psychological Science*, 19(11), 1095–1102. doi:10.1111/j.1467-9280.2008.02209.x.

M3 Wammes, J. D., Meade, M. E., and Fernandes, M. A. (2016). The drawing effect: Evidence for reliable and robust memory benefits in free recall. *Quarterly Journal of Experimental Psychology*, 69(9), 1752–1776. University of Waterloo repository record.

Dated energy statistics

D1 Ministry of New and Renewable Energy, Government of India. Physical Achievements, cumulative figures as at 30 September 2026. Values rounded from MW to GW; total renewable capacity includes large hydro.

D2 International Energy Agency. Global Energy Review 2026: Electricity supply. Global electricity-generation shares refer to 2025, not installed capacity or total primary energy.

Resources and solar energy

T1 U.S. Energy Information Administration (EIA). U.S. energy facts explained. Classification of primary, secondary, renewable and non-renewable energy.

T2 U.S. Bureau of Land Management. Oil Shale and Tar Sands. Resource definitions.

T3 Sandia National Laboratories / PV Performance Modeling Collaborative. Sun Position. Solar-position geometry; azimuth conventions must be stated.

T4 EIA. Solar thermal collectors. Flat plates and concentrating collectors.

T5 ENERGY STAR / U.S. Environmental Protection Agency. How solar water heaters work. Collector, circulation and heat-exchanger arrangements.

T6 Pennsylvania State University, EME 811 course materials. 4.2 Non-Imaging Concentrators. CPC acceptance and concentration.

T7 Food and Agriculture Organization (FAO). Small-Scale Postharvest Handling Practices: Processing of horticultural crops. Direct, indirect and mixed-mode solar dryers.

T8 U.S. Department of Energy (DOE). Solar Photovoltaic Cell Basics. Semiconductor conversion and cell technologies.

T9 DOE. Solar Photovoltaic Technology Basics. Cells, modules and system context.

References | Bioenergy, turbines, ocean and economics

Engineering sources continued

T10 DOE. Biopower: Energy, Heat, and Electricity. Combustion, gasification and power/heat pathways.

T11 FAO. Wood gas as engine fuel: Fuels for gasifiers. Moisture, ash, size and other feedstock properties.

T12 FAO. Biogas technology and plant designs. Fixed-dome and floating-drum arrangements.

T13 World Bank project environmental documentation (2010), biogas design annex. Fixed-dome and floating-drum plant construction and operating comparison.

T14 FAO. Wood gas as engine fuel: Types of gasifiers. Downdraft arrangement and operating trade-offs.

T15 IIT Delhi, gasifier research paper. Biomass gasification for rural electrification. Hot-zone tar reduction and producer-gas use.

T16 DOE. How Do Wind Turbines Work? Wind origin, lift, rotor types and power conversion.

T17 DOE. How a Wind Turbine Works. Nacelle, drivetrain, yaw, pitch and control components.

T18 DOE. How Hydropower Works. Plant layout and conversion sequence.

T19 DOE. Types of Hydropower Turbines. Impulse/reaction, Pelton, Francis and Kaplan.

T20 Hydro-Québec. Types of turbines. Runner configurations and resource suitability.

T21 Federal Energy Regulatory Commission. Hydropower Primer. Hydraulic structures, turbines and plant context.

T22 DOE. Enhanced Geothermal Systems. Heat, fluid, permeability and engineered reservoirs.

T23 DOE. Geothermal Electricity Generation. Dry-steam, flash and binary principles.

T24 EIA. Geothermal power plants. Steam, flash and binary system comparison.

T25 United Nations Framework Convention on Climate Change. Geothermal power plants. Generation technologies and applications.

T26 EIA. Ocean thermal energy conversion. Open/closed cycles and resource constraints.

T27 DOE. Marine Energy Glossary. OTEC, oscillating water columns and wave-converter motions.

T28 National Laboratory of the Rockies, System Advisor Model (SAM). Financial Metrics. Discounted cash flows and appraisal measures.

T29 SAM documentation. Financial Models Overview; SAM help manual (2020 edition). Analysis viewpoints and cash-flow conventions.

Use of sources. Dated figures are a revision snapshot and will change. General engineering principles are summarised for examination study; detailed design requires plant-specific calculations and standards. The memory schedule and short cues are practical learning aids, with the research limits stated on p.2.